

TABLE 151 **REST OF ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)**

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	2.10	2.49	2.94	3.48	4.11	4.88	5.80	6.92	18.7%
Academic and Research Institutes	3.03	3.52	4.07	4.72	5.47	6.36	7.41	8.66	16.3%
Other End Users	1.55	1.79	2.06	2.38	2.74	3.16	3.66	4.25	15.6%
Total	6.68	7.79	9.08	10.57	12.32	14.39	16.86	19.83	16.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.4.5 ASIA PACIFIC: IMPACT OF RECESSION

The impact of the recession on the global market would affect the Asia Pacific region as well. The Association of Southeast Asian Nations (ASEAN) projected a steady decline in the economic growth of Asia Pacific countries owing to the slower growth of their trading partners, namely, Europe, the US, and China. The overall GDP decline in the coming years is expected to affect the healthcare expenditure, spending, and wages in the industry with reduced import and export activities.

However, in Asia, countries focus on increasing their R&D in the pharmaceutical and biotechnology industry. India has seen a progression in terms of its R&D funding and rising government support. According to a report by India Brand Equity Foundation (IBEF), India's domestic pharmaceutical market was estimated at USD 42 billion in 2021. It is likely to reach USD 65 billion by 2024 and may further expand to reach USD 120–130 billion by 2030. Sun Pharma, a leading pharmaceutical company in India, announced that it intends to invest USD 600–650 million in R&D. On the other hand, Chinese pharmaceutical companies invested nearly 10% of their sales revenue in R&D in 2021.

For the overall Asia Pacific region, the impact of the recession is expected to be low during the forecast period. Japan's economy is expected to be more stable than other Asian countries. However, price inflation leading to lower consumption would lead to a slight decline in Japan's economic growth in 2023.

10.5 LATIN AMERICA

The Latin American microbiome sequencing services market is expected to reach USD 8.88 million by 2028 from USD 5.26 million in 2023, at a CAGR of 11.1% from 2023 to 2028.

The Latin American microbiome sequencing services market has been growing steadily in recent years. The region is witnessing an increased prevalence of chronic diseases, such as cancer, IBD, and diabetes, which have been linked to imbalances in the gut microbiome. As a result, there is a growing demand for microbiome sequencing services to better understand and manage these conditions. Additionally, the region's large and diverse population provides a unique opportunity for microbiome research and discovery.

The lack of awareness and understanding of the potential benefits of microbiome sequencing among healthcare providers and the general population is a key factor restraining the growth of this regional market. This has resulted in a relatively small market size and limited investment in the field. Additionally, the high cost of sequencing and the lack of standardization and regulation of microbiome testing can be a barrier to entry for smaller companies.

TABLE 152 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	2.01	2.23	2.47	2.74	3.04	3.37	3.76	4.20	11.1%
Rest of Latin America	2.28	2.52	2.79	3.08	3.41	3.78	4.20	4.69	11.0%
Total	4.29	4.75	5.26	5.82	6.44	7.15	7.96	8.88	11.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 153 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	1.54	1.70	1.88	2.08	2.30	2.55	2.83	3.16	10.9%
Sequencing and Library Preparation	2.09	2.33	2.60	2.90	3.23	3.61	4.04	4.54	11.8%
Data Analysis	0.66	0.72	0.78	0.85	0.92	1.00	1.09	1.19	8.7%
Total	4.29	4.75	5.26	5.82	6.44	7.15	7.96	8.88	11.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 154 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	1.77	1.97	2.20	2.45	2.72	3.04	3.40	3.82	11.7%
Whole-genome Sequencing	0.93	1.02	1.11	1.21	1.32	1.45	1.59	1.74	9.4%
Shotgun Sequencing	1.15	1.29	1.45	1.63	1.83	2.06	2.32	2.63	12.6%
Other Types	0.44	0.47	0.50	0.53	0.57	0.61	0.65	0.69	6.6%
Total	4.29	4.75	5.26	5.82	6.44	7.15	7.96	8.88	11.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 155 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	1.93	2.14	2.37	2.63	2.92	3.25	3.62	4.05	11.3%
Sequencing by Ligation	0.55	0.61	0.68	0.75	0.82	0.91	1.01	1.13	10.8%
Nanopore Sequencing	0.22	0.26	0.30	0.35	0.41	0.47	0.55	0.64	16.5%
Other Technologies	1.59	1.75	1.91	2.09	2.29	2.52	2.77	3.07	9.9%
Total	4.29	4.75	5.26	5.82	6.44	7.15	7.96	8.88	11.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 156 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	1.34	1.51	1.71	1.92	2.16	2.44	2.76	3.13	12.9%
Academic and Research Institutes	1.95	2.15	2.36	2.60	2.86	3.15	3.48	3.87	10.4%
Other End Users	1.00	1.09	1.19	1.31	1.43	1.56	1.72	1.89	9.6%
Total	4.29	4.75	5.26	5.82	6.44	7.15	7.96	8.88	11.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.5.1 BRAZIL

10.5.1.1 Growing adoption of next-generation sequencing technologies to propel growth

Brazil has been increasingly adopting NGS technologies for microbiome sequencing, which offer higher throughput, greater accuracy, and lower cost compared to traditional sequencing methods. This has led to the development of new microbiome sequencing services and the expansion of existing ones in the country.

The increasing prevalence of chronic diseases such as diabetes, obesity, and IBD is another major driver for market growth in Brazil, as it has created a high demand for microbiome sequencing services to better understand the role of the microbiome in disease development and progression.

TABLE 157 BRAZIL: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	0.72	0.80	0.89	0.98	1.08	1.20	1.34	1.49	11.0%
Sequencing and Library Preparation	0.98	1.10	1.22	1.36	1.52	1.70	1.91	2.14	11.9%
Data Analysis	0.31	0.34	0.37	0.40	0.43	0.47	0.51	0.56	8.9%
Total	2.01	2.23	2.47	2.74	3.04	3.37	3.76	4.20	11.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 158 BRAZIL: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	0.83	0.93	1.03	1.15	1.28	1.43	1.61	1.80	11.8%
Whole-genome Sequencing	0.44	0.48	0.52	0.57	0.62	0.68	0.75	0.83	9.5%
Shotgun Sequencing	0.54	0.61	0.68	0.77	0.86	0.97	1.09	1.24	12.7%
Other Types	0.21	0.22	0.24	0.25	0.27	0.29	0.31	0.33	6.9%
Total	2.01	2.23	2.47	2.74	3.04	3.37	3.76	4.20	11.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 159 BRAZIL: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	0.91	1.01	1.12	1.24	1.38	1.53	1.71	1.92	11.4%
Sequencing by Ligation	0.26	0.29	0.32	0.35	0.39	0.43	0.48	0.53	10.8%
Nanopore Sequencing	0.10	0.12	0.14	0.16	0.19	0.22	0.26	0.30	16.5%
Other Technologies	0.75	0.82	0.90	0.99	1.08	1.19	1.31	1.45	10.0%
Total	2.01	2.23	2.47	2.74	3.04	3.37	3.76	4.20	11.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 160 BRAZIL: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	0.63	0.71	0.80	0.90	1.02	1.15	1.30	1.47	12.9%
Academic and Research Institutes	0.91	1.01	1.11	1.22	1.35	1.49	1.65	1.83	10.5%
Other End Users	0.47	0.51	0.56	0.62	0.67	0.74	0.81	0.89	9.7%
Total	2.01	2.23	2.47	2.74	3.04	3.37	3.76	4.20	11.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.5.2 REST OF LATIN AMERICA

The Rest of Latin America mainly comprises Argentina, Colombia, Costa Rica, and Peru. The microbiome sequencing services market in the Rest of Latin America is estimated to reach USD 4.69 million by 2028 from USD 2.79 million in 2023, at a CAGR of 11.0% from 2023 to 2028.

Market growth in this region will be primarily driven by the awareness of and acceptance of personalized medicine, the increasing prevalence of chronic diseases (such as diabetes, cancer, and gastrointestinal disorders), and increasing investment in genomics research by governments and private organizations. However, certain regulatory barriers, such as complex approval processes, may create delays or difficulties in launching new microbiome sequencing services. Moreover, there may be a shortage of qualified professionals with the necessary expertise to perform and interpret microbiome sequencing.

TABLE 161 REST OF LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	0.82	0.90	1.00	1.10	1.21	1.35	1.49	1.66	10.8%
Sequencing and Library Preparation	1.11	1.24	1.38	1.53	1.71	1.91	2.14	2.40	11.8%
Data Analysis	0.35	0.38	0.41	0.45	0.48	0.53	0.57	0.62	8.5%
Total	2.28	2.52	2.79	3.08	3.41	3.78	4.20	4.69	11.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 162 REST OF LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	0.94	1.04	1.16	1.29	1.44	1.61	1.80	2.02	11.7%
Whole-genome Sequencing	0.50	0.54	0.59	0.64	0.70	0.76	0.84	0.92	9.2%
Shotgun Sequencing	0.61	0.68	0.77	0.86	0.97	1.09	1.23	1.39	12.6%
Other Types	0.23	0.25	0.27	0.28	0.30	0.32	0.34	0.36	6.4%
Total	2.28	2.52	2.79	3.08	3.41	3.78	4.20	4.69	11.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 163 REST OF LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	1.02	1.13	1.26	1.39	1.54	1.71	1.91	2.13	11.2%
Sequencing by Ligation	0.29	0.32	0.36	0.39	0.44	0.48	0.53	0.60	10.7%
Nanopore Sequencing	0.12	0.14	0.16	0.19	0.22	0.25	0.29	0.34	16.6%
Other Technologies	0.84	0.93	1.01	1.11	1.21	1.33	1.46	1.61	9.8%
Total	2.28	2.52	2.79	3.08	3.41	3.78	4.20	4.69	11.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 164 REST OF LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	0.71	0.80	0.90	1.02	1.14	1.29	1.46	1.66	12.9%
Academic and Research Institutes	1.03	1.14	1.25	1.37	1.51	1.66	1.84	2.04	10.3%
Other End Users	0.53	0.58	0.63	0.69	0.75	0.82	0.90	0.99	9.5%
Total	2.28	2.52	2.79	3.08	3.41	3.78	4.20	4.69	11.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.5.3 LATIN AMERICA: IMPACT OF RECESSION

A US recession would affect Latin America's real GDP outlook by impacting trade and the overall economy. GDP growth of 3.3% in 2022 is expected to slow by 1.5% in 2023 in Latin America. However, government bodies and pharmaceutical companies are expanding the pharmaceutical and biotechnology industry in countries such as Brazil. In October 2022, Luis Inácio Lula da Silva, one of the candidates in the presidential election and winner of Brazil's first-round elections, pledged to increase healthcare spending by investing in the Unified Health System (SUS) and health infrastructure, thereby expanding the domestic pharmaceutical industry. In November 2022, MedQuímica Indústria Farmacêutica (MedQuímica), a wholly owned subsidiary of Lupin in Brazil, signed a definitive agreement to acquire all rights to nine medicines from BL Indústria Ótica Ltda., a subsidiary of Bausch Health Companies Inc., to enhance its product portfolio in Brazil. Such initiatives are likely to encourage the growth of the pharmaceutical and biotechnology industry in 2023 and 2024.

Inflation has reached an inflection point in most countries in Latin America with tighter monetary conditions, stabilizing prices for energy, health and food commodities, transportation costs, and supply chain disruptions. All these factors would create an impact on the pharmaceutical and biotechnology industry and related technologies, such as metagenomic sequencing. Market growth is expected to slow in 2023-24 and will eventually start to stabilize and recover in 2025-26.

10.6 MIDDLE EAST & AFRICA

The microbiome sequencing services market in the Middle East & Africa is projected to reach USD 5.55 million by 2028 from USD 3.55 million in 2023, at a CAGR of 9.3% from 2023 to 2028.

The increasing awareness and understanding of the importance of the microbiome in human health and diseases are contributing to the growth of this regional market. However, the market growth is likely to be restrained by factors such as the high costs associated with microbiome sequencing services and a lack of skilled professionals in the region. Despite these challenges, there are several opportunities for market growth in MEA, including the potential for increased investment in research and development, partnerships and collaborations between academic and industrial sectors, and the development of novel technologies and analytical tools.

TABLE 165 MIDDLE EAST & AFRICA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	1.06	1.16	1.27	1.38	1.51	1.65	1.80	1.97	9.2%
Sequencing and Library Preparation	1.45	1.60	1.76	1.93	2.13	2.34	2.58	2.85	10.2%
Data Analysis	0.46	0.49	0.53	0.56	0.60	0.64	0.68	0.73	6.8%
Total	2.97	3.25	3.55	3.88	4.23	4.63	5.06	5.55	9.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 166 MIDDLE EAST & AFRICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	1.22	1.35	1.48	1.63	1.79	1.97	2.17	2.40	10.1%
Whole-genome Sequencing	0.65	0.70	0.75	0.81	0.87	0.93	1.00	1.08	7.5%
Shotgun Sequencing	0.79	0.88	0.98	1.09	1.21	1.34	1.48	1.65	11.0%
Other Types	0.31	0.32	0.34	0.35	0.37	0.39	0.40	0.42	4.6%
Total	2.97	3.25	3.55	3.88	4.23	4.63	5.06	5.55	9.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis